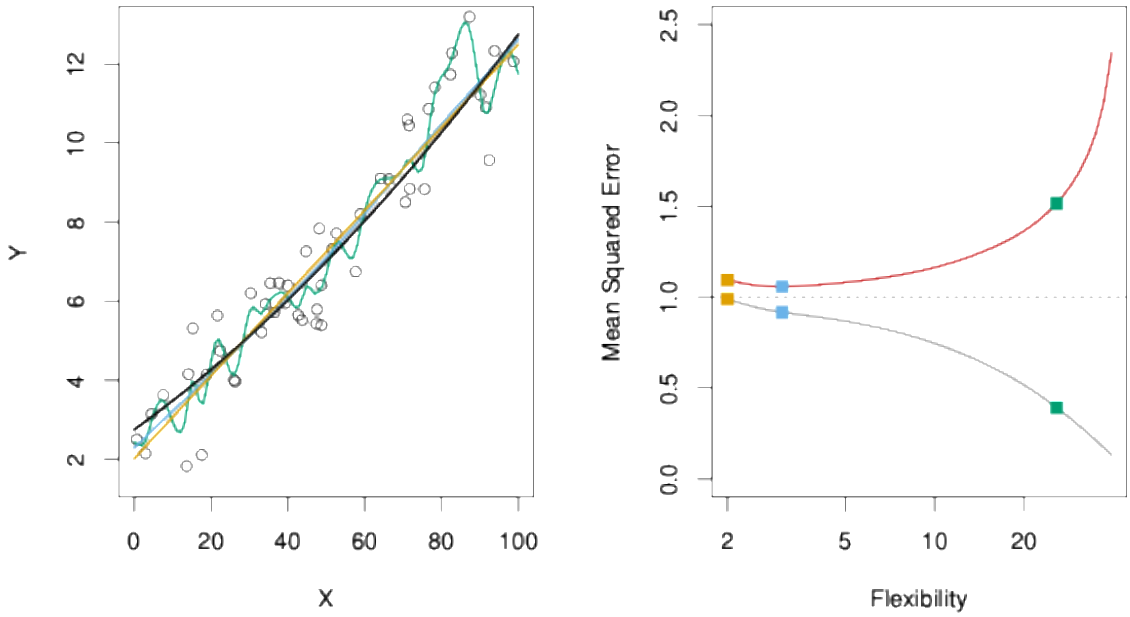
DS 6021: Intro to Predictive Modeling
Quiz #3: October 23, 2026

Name: **Quiz #3 Solutions**

Quiz Format

This is a paper and pen/pencil quiz. It will be closed book. No reference materials of any kind will be allowed. You will have **30 minutes** to answer the questions below. A few sentences should be fine for each, and bullet points are also acceptable. The quiz is worth a total of **20 points**.

Question 1: Bias-Variance Tradeoff [5 points]



The plot above is from Introduction to Statistical Learning and was reviewed in class. In the plot on the right, the red line is the test (out-of-sample) MSE and the grey line is the training RMSE. The colored squares on the right are the calculated train and test RMSE that correspond to the curves on the left. (e.g. The green squares on the right correspond to the train and test RMSE for the green curve on the left.)

Using this plot answer the following questions:

a. [2 pts] Does the green line on the plot on the left demonstrate overfitting or underfitting?

Solution: The green line demonstrates **overfitting**. It has the lowest training MSE, but the highest test MSE, both of which are hallmarks of high variance/overfitting. Also, we can visually see from the plot on the left that it adhered quite closely to the training data.

b. [3 pts] Looking at both plots, which model would you say is the best and why?

Solution: There are two defensible answers to this question, and as long as you provided a clear rationale you can get credit for either answer.

- **Option 1**: The blue model is the best. Rationale: It has the lowest test MSE, indicating the right balance between bias and variance. Additionally, its train and test MSE are close together, and both are near the dashed line (irreducible error), so adding complexity to the model would not give you much.
- **Option 2**: The orange model because it is the simplest of the three, and its training and test error curves are nearly equal to the blue. Depending on the model in question, you might prefer something simpler, and so orange could be a defensible choice.

Question 2: Polynomial and Spline Regression [4 points]

a. [2 pts] What is a key difference between polynomial and spline regression? There is more than one difference, but you just need to name one for full credit.

Solution: Any one of the following answers will receive full credit:

- **Global vs local fitting**: Polynomial regression fits one high-degree polynomial over the whole range of X, whereas splines fit separate lower degree polynomials on intervals between knots.
- **Behavior at the boundaries**: High degree polynomials can behave erratically at the boundaries of the data, whereas the flexibility of splines means they are more stable at the boundaries, especially with natural cubic splines.
- **Sensitive to outliers/changes in data**: Because polynomials fit one polynomial over the whole range of X, changing one data point can impact the fit everywhere. Splines adapt more locally.

b. [2 pts] Why are these still considered linear models, even though they introduce higher order terms (e.g. X^2, X^3, etc.)

Solution: These models are linear in the coefficients (βs). The X^2, X^3, and spline basis functions are new features computed from X, and can still be written as a linear combination of the βs.

$$Y = \beta_0 + \sum_{j=1}^{d} \beta_j X^j + \epsilon$$

Question 3: Sklearn [5 points]

Examine the following Python code:

```
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline

numeric_columns = ["flipper_length_mm"]
```

```
categorical_columns = ["sex", "species"]

preprocess = ColumnTransformer([
    ("num", StandardScaler(), numeric_columns),
    ("cat", OneHotEncoder(drop="first"), categorical_columns),
])

pipe = Pipeline([
    ("preprocess", preprocess),
    ("model", LinearRegression()),
])

pipe.fit(X_train, y_train)
```

Briefly explain what each pipeline step does.

Solution (one point for correctly identifying each of the following):

- StandardScaler(): Takes the numeric column () and scales it to 0 mean and SD 1.
- OneHotEncoder(drop="first") on sex and species takes each category and converts it to 0/1 and drops one column per variable. Each dropped level becomes the reference category.
- ColumnTransformer applies different preprocessing to different columns as described in the above two bullet points
- LinearRegression fits an OLS model on the transformed features to fit y
- pipe.fit runs the above steps in order. It preprocesses the data (scaling and encoding by column type, via the ColumnTransformer) and then fits the regression.

Question 4: Regularization [6 points]

Ridge, Lasso, and Elastic Net are forms of regularization.

a. [3pts] What common goal do they share?

Solution: All three control overfitting by shrinking coefficients and improving generalization and stability. Approximately 1.5 points will be given for including each of the following:

- Reduce overfitting by **adding a penalty term** to the loss function so you minimize RSS + λ•penalty
- **Shrinking coefficients towards zero**, which gives an increase in bias for a drop in variance.

b. [3pts] How do Ridge and Lasso differ in how they affect coefficients?

Solution: Ridge regression shrinks coefficients toward zero but rarely shrinks them to exactly zero, so every feature stays in the model. Lasso can shrink some coefficients exactly to zero, which performs automatic feature selection. If you correctly write the equations for ridge and lasso but don't discuss this distinction, you will only receive partial credit.